1. Safe reinforcement learning with mixture density network, with application to autonomous driving
   Abstrak : This paper presents a safe reinforcement learning system for automated driving that benefits from multimodal future trajectory predictions. We propose a safety system that consists of two safety components: a rule-based and a multimodal learning-based safety system. The rule-based module is based on common driving rules. On the other hand, the multi-modal learning-based safety module is a data-driven safety rule that learns safety patterns from historical driving data. Specifically, it utilizes mixture density recurrent neural networks (MD-RNN) for multimodal future trajectory predictions to mimic the potential behaviors of an autonomous agent and consequently accelerate the learning process. Our simulation results demonstrate that the proposed safety system outperforms previously reported results in terms of average reward and collision frequency.

2. Advancing autonomous driving system testing: Demands, challenges, and future directions
   Abstrak : Context:
   Autonomous driving systems (ADSs) promise enhanced transportation efficiency but face critical challenges in ensuring reliability across complex driving environments. Effective testing is essential to validate ADS performance and mitigate real-world risks.
   Objective:
   This study investigates current ADS testing practices for both modular and end-to-end systems, identifies key demands (needs required by practitioners and researchers), and examines gaps between research and real-world demands. We review critical testing techniques and extend to involve Vehicle-to-Everything (V2X) communication and Foundation Models (FMs), including large language models and vision foundation models, in enhancing ADS testing performance. We provide literature reviews and outline future directions for each demand of industry practitioners and academic researchers.
   Methods:
   A large-scale survey was conducted with 100 participants, including industry practitioners and academic researchers. We first discuss survey questions with professionals, distribute them to industry practitioners and academic researchers, and conduct follow-ups. Quantitative and qualitative analyses uncover key trends and challenges.
   Results:

Findings reveal that existing ADS testing techniques struggle to evaluate real-world performance comprehensively, including the diversity of corner cases, the gap between simulation and real-world testing, the lack of comprehensive testing criteria, potential attacks, practical deployment for V2X, and the computational costs for FMs. By analyzing participants' responses and 105 relevant papers, we summarize the future research directions.
Conclusion:
Our study highlights critical research gaps in ADS testing and underscores the demands of industry practitioners and academic researchers. We provide future directions for ADS: comprehensive testing criteria, cross-model collaboration in V2X, enhancing cross-modality (e.g., text and image) adaptation in FM testing, and scalable ADS validation frameworks. These insights contribute to advancing software engineering practices for ADS development, ensuring safer and more reliable autonomous systems.

3. An Improved Safe Soft Actor-Critic Path Planning Algorithm for Autonomous Vehicles Based on a Dual-Stream Q-Network and Dynamic Analytic Hierarchy Process
   Abstrak : To address the conflict between navigation performance and safety constraints in safe reinforcement learning, this paper proposes Dual Stream-Analytic Hierarchy Process-Safe Soft Actor (DS-AHP-SAC), a safe soft actor-critic algorithm based on a dual-stream Q-network and dynamic Analytic Hierarchy Process (AHP) stratified experience replay. The algorithm achieves a balance between reward maximization and constraint satisfaction through three synergistic designs: (1) decoupling the Q-network into independent navigation and safety value streams to eliminate gradient interference at the Critic level and mitigate gradient competition at the Actor level; (2) constructing a three-criterion dynamic sampling strategy based on AHP, incorporating safety urgency, information value, and scarcity to enable phase-adaptive experience replay; (3) designing a curriculum-scheduling scheme that linearly increases the safety constraint weight during training, preventing policy degradation caused by premature imposition of high safety penalties. Experimental results in a two-dimensional continuous navigation environment demonstrate that DS-AHP-SAC reduces the violation rate by 20.0% compared to unconstrained SAC without sacrificing navigation success rate, while avoiding the policy collapse observed in SAC-Lagrangian. Ablation studies validate the necessity of the dual-stream Q-network and the convergence acceleration effect of dynamic AHP.

4. 4 - Artificial intelligence for autonomous vehicle safety
   Abstrak : The integration of Artificial Intelligence (AI) into autonomous vehicle (AV) systems marks a significant shift in automotive technology, enhancing road

safety dramatically. This chapter explores AI's critical role in improving AV safety mechanisms as the industry moves toward full autonomy. The use of sophisticated AI models is essential for navigating complex real-world driving scenarios with exceptional precision and reliability. Central to this evolution are advanced deep learning techniques such as convolutional neural networks (CNNs) for image processing and object detection, and recurrent neural networks (RNNs) for analyzing temporal data and predicting movements. Additionally, deep reinforcement learning (DRL) enables AVs to make instantaneous decisions in dynamic environments, simulating human decision-making processes. More recently, Transformers have gained significant attention and have demonstrated powerful capabilities across a wide range of applications for autonomous driving. Despite substantial advancements, the application of AI in AVs faces challenges including the need for extensive testing, development of universal safety standards, and integration with existing infrastructure. However, the potential for AI to transform AV safety is immense. Future research should focus on developing hybrid AI models that merge CNNs, RNNs, and DRL for a better environmental understanding, devising AI systems for ethical reasoning, and ensuring transparency and accountability in AI decisions. Effective integration of AI into AV safety systems can make autonomous driving safer, but achieving this requires overcoming existing hurdles through innovative research and interdisciplinary cooperation. As AI technology advances, it will increasingly influence the future of autonomous driving and vehicular safety.

5. Enhancing autonomous driving safety in real lane-changing scenarios under friction variability: A friction-adaptive shield reinforcement learning framework
Abstrak : How autonomous vehicles (AVs) can learn safety-aware policies in real-world lane-changing scenarios under friction variability remains underexplored. To address this, a friction-adaptive shield reinforcement learning (FA-SRL) framework is proposed in this study. FA-SRL incorporates a friction-adaptive shield that actively intervenes to address the impact of variable friction on vehicle dynamics through a generalized time-to-collision (GTTC)-based risk assessment and graded risk rewards. Simulation scenarios based on real-world datasets with varying traffic densities are constructed to rigorously train and evaluate FA-SRL variants against their baselines (PPO, SAC, TD3) and naturalistic driving behavior. Training results demonstrate that FA-SRL significantly enhances final rewards and convergence speed, optimizing by 11.31 % and 34.76 %, respectively, compared to baselines. Evaluations indicate substantial safety improvements with FA-SRL variants; particularly, FA-SRL (SAC) achieves the optimal and most balanced performance with success rates of 100 % in low-density and 97.33 % in high-density traffic and statistical safety improvements

compared to human driving under varying traffic densities. Specifically, the safe time-step ratio in low-density and high-density scenarios is improved by 1.28 % and 2.97 %, respectively. Moreover, the efficiency performance of learned policies significantly outperforms human driving behavior, with an average reduction of 13.63 % in task completion time. However, the conservative friction-adaptive shield in FA-SRL slightly compromises efficiency relative to baselines. These findings validate the effectiveness and robustness of the proposed FA-SRL under inevitable friction variability, highlight its potential for deployment in safety–critical autonomous driving requiring a careful balance between safety and efficiency, and further suggest its broader value as a principled framework for embedding physics-based safety constraints into policy learning.

6. ADAS-RL: Safety learning approach for stable autonomous driving
   Abstrak :
   Stability is the most significant component of an autonomous driving system, affecting both the lives of drivers and pedestrians and traffic flow. Reinforcement learning (RL) is a representative technology used in autonomous driving, but it has challenges because it is based on trial and error. In this letter, we propose an efficient learning approach for stable autonomous driving. The proposed deep reinforcement learning based approach can address the partially observable scenario in mixed traffic which includes both autonomous vehicles and human-driven vehicles. Simulation results show that the proposed model outperforms the control-theoretic and vanilla RL approaches. Furthermore, we confirm the effect of the sync-penalty, which teaches the agent about unsafe decisions without experiencing the accidents.

7. Balancing driving safety and performance in reinforcement learning via adaptive expert guidance and regional policy optimization
   Abstrak :
   Safety remains a critical challenge when training practical driving policy with reinforcement learning (RL). Existing safe RL approaches often struggle to balance exploratory behavior with strict safety constraints, leading to overly conservative or unsafe policies. This work presents a novel safe RL framework that leverages a dual risk perception module to optimize the ensemble soft mixture-of-experts policy via feasible policy iteration, which can achieve efficient, safe, and stable policy learning. The dual risk perception module jointly quantifies knowledge-based uncertainty by an ensemble performance function, and evaluates long-term safety risks via a pre-trained constraint decay function. Then, the policy fuses ensemble learning with differentiable soft routing mechanism to generate diverse and adaptive candidate actions under controllable computational complexity. Finally, a regional optimization

mechanism maximizes policy performance within feasible regions and enforces safety recovery outside. During policy inference, a two-stage safety discrimination mechanism further ensures robust decision and control. Experiments in the highway-env benchmark show that the proposed method significantly outperforms multiple mainstream safe RL baselines in return, success rate and safety, which demonstrates its effectiveness in achieving a superior balance between performance and safety.

8. Safe MBPO-based speed and lane change method integrating decision-making and control for autonomous vehicles
   Abstrak :
   In this paper, a new reinforcement learning method based on safe model-based policy optimization (SMBPO) that integrates decision-making and control problem with hierarchical action space of autonomous driving is proposed. This method uses interaction experiences to build an environment model that interacts with the agent and ensures efficient balance between safety and driving efficiency by incorporating the safe state set, terminal cost, and integrating them into the safety policy optimization process. A hierarchical reinforcement learning architecture is designed to decompose the action space into continuous longitudinal control and discrete lateral maneuvers, enabling closed-loop integration of decisions and motion execution. For this decision-making and control problem, the goal is to achieve optimal safety, while also considering driving efficiency of the autonomous vehicle, which is embodied in maximizing the reward functions of the relevant aspects in reinforcement learning algorithms. The SMBPO algorithm implements safety constraints through Lagrangian relaxation, dynamically adjusting the trade-off between reward maximization and hard safety requirements such as collision avoidance. Specifically, a dual-layer safety check mechanism validates both lateral safe boundary compliance and longitudinal thresholds during model rollouts, terminating unsafe trajectories in real-time. Besides, the generalization tests and comparative experiments with other classical reinforcement learning algorithms are carried out. In comparison with the benchmark methods, our approach demonstrates approximately 40 % of their sample efficiency while achieving higher performance metrics such as safety rates close to 100 % and 85 % in the two modes respectively.

9. AI-driven path planning for autonomous vehicles: A review of algorithms, optimization, and cooperative strategies
   Abstrak :
   Autonomous vehicle (AV) path planning is a core decision-making function that ensures safe, efficient, and comfortable navigation in dynamic traffic

environments. This review synthesizes recent advances in AI-driven path planning by systematically categorizing algorithms into graph-based, sampling-based, optimization-based, learning-based, and hybrid frameworks. Key performance objectives, including collision avoidance, ride comfort, energy efficiency, and real-time feasibility are discussed alongside representative methods such as A*, D*, RRT/RRT**, model predictive control, heuristic/metaheuristic optimizers, deep reinforcement learning, and imitation learning. The review further analyzes cooperative path planning enabled by V2X connectivity, highlighting coordination strategies for platooning, merging, and intersection negotiation. Challenges including uncertainty handling, generalization across scenarios, computational constraints, safety assurance, and sim-to-real transfer are critically examined. Finally, open research directions are presented, emphasizing unified evaluation benchmarks, interpretable learning-based planners, multi-agent safety guarantees, and robust planning under imperfect perception. The paper provides an updated reference framework for researchers and practitioners seeking scalable and deployable path planning solutions for next-generation AV systems.

10. Nexus of control: A dynamic consensus framework for energy-positive autonomous emergency braking

    Abstrak :

    Automatic emergency braking (AEB) in electric vehicles must dissipate kinetic energy rapidly while preserving collision safety, yet conventional AEB logic often prioritizes friction braking and leaves regenerative capacity underused. This limitation becomes more difficult in cooperative vehicle-to-everything (V2X) scenarios, where shared braking intent can improve fleet-level energy recovery but faulty or malicious messages may corrupt the collective state estimate. This paper presents Consensus-Driven Predictive Energy Management (CoPEM), a closed-loop framework for energy-positive AEB under adversarial cooperative information. CoPEM couples four modules: a trust-weighted dynamic consensus layer for bounded-error state agreement, an Eco-TES Transformer for predicting battery safe operating boundaries, a multi-objective soft actor-critic agent for regenerative-friction brake blending, and a high-order control barrier function (HOCBF) safety filter that projects nominal actions onto a collision-safe set. The framework is evaluated in software-in-the-loop simulations covering Euro NCAP-inspired vehicle-to-vehicle braking, obstructed vulnerable-road-user scenarios, and six-vehicle cooperative emergency braking with stochastic V2X latency, packet loss, and Byzantine fault injection. CoPEM improves average regenerative energy recovery by 36.5% in single-vehicle AEB and by 187.9% at the fleet level relative to a non-cooperative baseline under matched initial conditions. Under the boundary stress case ($f = 2$, $N = 6$), corresponding to $f/N = 1/3$, effective V2X

consensus quality decreases from 98.5% to 29.6%, while the controller shifts toward local-sensing HOCBF fallback and maintains low simulated collision risk. These results support CoPEM as a simulation-validated architecture for trust-aware, battery-aware, and formally filtered energy-positive emergency braking, while motivating subsequent HIL and real-vehicle validation before deployment-level claims are made.

11. Velocity adaptive preview path tracking control for autonomous vehicles using deep reinforcement learning with Lyapunov function-based reward shaping
    Abstrak :
    Traditional rule-based path tracking controllers of autonomous vehicles (AVs) often struggle with nonlinear vehicle dynamics, parameter tuning, and changes in road conditions. Learning-based methods partly address these issues, but remain limited in adapting to velocity changes, exploiting the preview information from varying-curvature roads, and physically interpretable reward shaping that encourages Lyapunov function decrease. To address these challenges, a velocity adaptive preview soft actor–critic (APSAC) algorithm based on Lyapunov function-based reward shaping is proposed in this paper. Firstly, using a multi-point preview-based physical reference model for interpretability, longitudinal velocity is incorporated into the APSAC policy state and coordinates velocity adaptive preview-state construction, multiple reward parameters, and training with continuous acceleration and deceleration. Secondly, a basic reward mechanism is designed, which integrates state-dependent dense rewards with trigger-based rewards that drive state convergence. Building on this basic reward, a Lyapunov function-based potential-difference term is added to form the velocity adaptive hybrid reward. Under the stated assumptions, the conditional relationship between Lyapunov potential descent and basic reward maximization is examined. The convergence of idealized learning iterations is analyzed. Finally, simulation experiments and semi-physical hardware-in-the-loop (HiL) experiments demonstrate the superiority and engineering feasibility of the proposed method.

12. A comprehensive survey on deep reinforcement learning for autonomous vehicles urban driving: Approaches, challenges, and future directions
    Abstrak :
    Deep Reinforcement Learning (DRL) integration has resulted in notable progress in the development of autonomous driving systems (ADS). DRL is an effective approach for autonomous vehicles (AVs) because of its capacity to manage intricate decision-making processes in dynamic and unpredictable contexts. This study provides a comprehensive review of DRL approaches applied to ADS, with a focus on key components such as planning, control, localization,

mapping, and scene understanding. We examine various DRL methods: value-based, policy-based, and actor-critic, and highlight their contributions to real-time decision-making, low-level vehicle control, path planning, and trajectory optimization. In addition, we discuss advanced DRL extensions, such as imitation learning (IL), multi-agent systems, and safe DRL, which address important challenges related to safety, efficiency, and robustness in autonomous systems. This study also reviews modern simulation environments that play a crucial role in testing and training DRL models within virtual settings. Finally, we outline future research directions, emphasizing the need for improved generalization, sample efficiency, and ethical considerations in DRL-based ADS. The purpose of this survey is to provide insights that can contribute to the development of safer and more efficient AVs capable of operating under diverse and challenging conditions.

13. **A Novel Mountain Driving Unity Simulated Environment for Autonomous Vehicles**

    Abstrak :

    The simulated driving environment provides a low cost and time-saving platform to test the performance of the autonomous vehicle by linkage with existing machine learning approaches. However, most of existing simulated driving environments focus on building flat roads in urban areas. Still, they neglected to endeavour the tough steep, curvy hill roads, such as mountain paths around suburban areas. In this study, by deploying in Unity engine, we developed the first complex mountain driving simulated environment with characterizing continuous curves and up/downhill. Then, two state-of-art reinforcement learning (RL) algorithms are used to train a vehicle agent and test the performance of autonomous vehicles in our developed simulated environment. Also, we set 5 different levels of vehicle's speeds and observe the cumulative rewards during the vehicle agent training. Our demonstration presents the developed environment supports for complex mountain scenario configurations and RL-based autonomous vehicles, and our findings show that the vehicle agent could achieve high cumulative rewards during the training stage, suggesting that our work is a potential new simulation environment for autonomous vehicles research. The demonstration video can be viewed via the link: https://youtu.be/0wSqGeCn-NU.

14. **Ethical Autonomous Driving Systems in Mixed-Traffic Environments: Architecture Definition and Regional Viability Evaluation**

    Abstrak :

    In mixed-traffic environments, autonomous driving systems (ADSs) interact with humans and need to exhibit ethical behavior that considers stakeholder

satisfaction and contributes to public safety. From a human–systems integration perspective, this study defines an ethical ADS architecture aligned with region-specific ethical values and evaluates its regional viability. Drawing on P. F. Strawson's concept of reactive attitudes, ethical ADSs exhibit collaborative ethical behavior by creating safety zones in cooperation with surrounding vehicles for safe pedestrian crossings. Safety zone creation can promote public safety awareness diffusion but affects traffic efficiency. Viability is evaluated across urban, suburban, and rural scenarios using two probabilistic indicators: expected regional traffic delay rate (ERTDR) and time for community diffusion of public safety awareness (TCD). Allowable deceleration is used as a control parameter for non-emergency safety zone creation, and the balance between ERTDR and TCD is evaluated for each scenario. Results indicate relatively high urban viability, conditional suburban viability, and insufficient rural viability without complementary measures. Robustness analysis of the results shows that TCD is particularly sensitive to the diffusion coefficient of public safety awareness, while the principal finding that the rural scenario has the longest TCD is retained under parameter uncertainty and an alternative diffusion model.

15. A new model for autonomous driving in mountainous terrain with reinforcement learning using Unity

    Abstrak :

    Self-driving cars in mountainous terrain have many critical applications. For example, autonomous off-road vehicles for rescue and exploration, self-driving trucks for the mining industries, and autonomous shuttle services for remote resorts. Training autonomous vehicles in these environments has many challenges compared to urban or highway environments. These challenges include steep slopes, winding roads, and off-road obstacles. This research presents a new reinforcement learning model to drive in this complex environment. The model explores the effect of using fewer sensors for performance enhancement and cost-effectiveness in this environment. These sensors are RADAR, IMU, and GPS. Unity Engine with ML Agent is used as a simulator for training, as it provides a virtual environment for efficient and safe driving. The selected sensors facilitate a seamless knowledge transfer from the simulator to a real vehicle. They also provide sufficient information for measuring road gradient and angle between successive lane segments. A new reward system is designed based on the Proximal Policy Optimisation (PPO) algorithm with Behavioural Cloning (BC). The experimental findings suggest that the proposed model can navigate mountainous terrain safely and efficiently while avoiding collisions and staying in the correct lane. The model achieved a low total collision rate of approximately 0.04 collisions per obstacle, a lane exit rate

of 0.014, maintained an average speed of 22 km/h, and a maximum speed of 48 km/h in complex simulated environments.

16. **A Safe and Efficient Navigation Framework for Ground Vehicles on Uneven Terrain Considering Kinematic Constraints and Terrain Traversability**

Abstrak :
Ground vehicles navigating uneven terrain must simultaneously guarantee motion safety and efficiency. Safety requires that the planned waypoints lie in highly traversable terrain, while ensuring vehicle reachability to these waypoints, which must be kinematically feasible. Efficiency demands fewer detours and smoother paths that avoid excessive vehicle acceleration and steering. However, existing path planning research for uneven terrain fails to comprehensively integrate vehicle kinematic constraints, terrain factors, path smoothness, rollover risk, and total path length. To address this problem, this paper proposes a novel navigation framework. It first integrates terrain slope, flatness, elevation variation, and sparsity to generate a 2D global terrain traversability cost map. Subsequently, a three-phase path planning algorithm integrates A*, guided Rapidly-exploring Random Tree (RRT), and our proposed Kinematic and Terrain-Aware Probabilistic Roadmap (KT-PRM) local re-planning algorithm, which jointly considers multiple factors including ground vehicle kinematic constraints, terrain factors, path smoothness, rollover risk, and path length. This three-phase combination delivers safe, smooth, and short global paths over uneven terrain within a relatively short planning time. Finally, Nonlinear Model Predictive Control (NMPC) is employed for path tracking in the framework. Experiments were conducted in both simulated and real-world uneven terrain environments. The results demonstrated that the three-phase path planning algorithm integrated with our proposed KT-PRM algorithm achieves comprehensive performance in generating safer, smoother, and shorter paths. Our proposed navigation framework achieves safer and more efficient navigation compared with existing navigation frameworks.

17. **Hierarchical Terrain-Aware Navigation via Bayesian-Guided RRT* and Flat-Corridor CiLQR**

Abstrak :
Autonomous navigation in unstructured, off-road environments remains a critical challenge due to complex terrain and limited prior knowledge. This letter presents a hierarchical framework to generate paths across locally flat regions and optimize spatio-temporal trajectories under vehicle kinodynamics. The first step employs a terrain-aware RRT* that integrates Bayesian-learning-guided sampling with a directional reachability index, enabling efficient exploration in complex environments while avoiding dead-end traps. The resulting path

explicitly accounts for local roughness and slope. The second step applies a constrained iterative linear quadratic regulator (CiLQR) trajectory optimizer that refines the path within a dynamic flat-corridor, where terrain risk is embedded into the optimization. The optimizer enforces vehicle stability constraints and adaptively regulates speed based on terrain risk and curvature, yielding dynamically feasible trajectory. Extensive simulations in off-road scenarios demonstrate that the proposed framework outperforms state-of-the-art methods, producing feasible trajectories with up to 34 % lower vehicle jolt. Furthermore, real-world tests on a 40-ton articulated truck in the Alxa Desert validate the framework's robustness and practical effectiveness.

18. **Global Path Planning Methods Based on the Relationship Between Traversability Capability and Terrain Matching**

    Abstrak :

    In contrast to structured urban settings, road networks in post-disaster or unstructured wildland environments are often incomplete or compromised. Navigation in these contexts requires navigating complex terrains and mitigating potential hazards that impede unmanned ground vehicles (UGVs). While high-mobility off-road vehicles are specifically designed to traverse challenging features like ditches and steep slopes, traditional path planning algorithms often fail to exploit these capabilities. These algorithms typically suffer from a binary focus, either relying strictly on road networks or ignoring them altogether, thereby neglecting the synergy between infrastructure and vehicle mobility. This chapter introduces a global path planning method based on traversability analysis and terrain matching to bridge this gap. The methodology incorporates a grid-based traversability evaluation, a road network expansion algorithm for densifying critical segments, and a unified planning strategy. By correlating terrain characteristics with vehicle mobility limits and optimizing the road network density, the proposed framework achieves an integrated on-road and off-road planning solution that maximizes the operational efficiency of high-mobility vehicles in degraded environments.

19. **A Ground Segmentation Algorithm for Slopes**

    Abstrak :

    When autonomous vehicles are driving in complex environments, the presence of uneven ground and obstacles can impact the safety of the vehicle. Traditional obstacle detection algorithms overlook the impact of sloped terrain on vehicle path planning. In environmental perception, determining whether a slope is passable has become a hot and challenging issue. This paper proposes a 3D LiDAR-based ground segmentation algorithm. This method relies on data obtained from experimental equipment for terrain information collection

systems ahead of the vehicle, enabling it to distinguish between ground points and non-ground points in steep terrain. Experiments show that this method achieves good results in both static and dynamic slope detection.

20. **Automated Driving System Challenges in Rural Appalachia**

Abstrak :
The development of autonomous driving technology has predominantly focused on urban and suburban areas. Deployment of automated driving systems in regions such as rural Appalachia present unique challenges such as narrow and winding roads and degradation of localization. Scenarios in rural Appalachia that required manual intervention by a driver during autonomous driving experiments were investigated across three unique routes. The research identified the technological and environmental limitations that contributed to these interventions and how they may differ from urban settings. The goal was to provide insights into the factors that hinder autonomous vehicle performance in rural areas and guide the development of more adaptable and robust systems capable of operating reliably in diverse environments, extending the benefits of autonomous driving to rural populations and ensuring equitable access to advancements in transportation. Driving experiments resulted in 1,884 total interventions and revealed trends in the reasons and locations for intervention across the three routes. In rural areas the leading causes of takeover were localization issues, accounting for 30.4% of total events, environmental traffic uncertainties, responsible for 20.3%, and object detection challenges, comprising 15.2%. Whereas urban settings saw roundabouts, environmental traffic uncertainties, and stoplight detection errors as the most common reasons with respective percentages of 19.5%, 17.7%, and 15.4%, revealing key differences between environments.

21. **PROXIMAL POLICY OPTIMIZATION FOR COLLISION AVOIDANCE AND MOTION PLANNING IN AUTONOMOUS VEHICLES: A MATHEMATICAL MODELING PERSPECTIVE**

Abstrak :
Motion planning and collision avoidance (MPCA) are critical in autonomous driving systems (ADS). Current deep reinforcement learning (DRL) ADS utilize discrete action space algorithms for MPCA, leading to inefficient and inaccurate MPCA in complex driving environments. This paper focuses on advanced mathematical modeling techniques in the context of autonomous vehicle (AV) dynamics and proposes an end-to-end proximal policy optimization (PPO)-based DRL framework for MPCA in complex environments. The proposed framework comprises three main components. First, a 3D virtual environment

and a model car are designed, equipped with multimodality sensors including Cameras, LIDAR, and GPS. Second, an immediate reward system is designed with varied control bonds and a convolutional neural network (CNN)-based model for feature extraction and curve learning. The model is optimized using PPO in a Markov decision style of direct perception and driving. Finally, the network is embedded in the 3D model car for a real test drive. The framework is evaluated through various experiments, including a detailed ablation of driving in real scenarios. The results demonstrate that our framework allows AVs to perform efficient and accurate MPCA in a complex virtual environment, and it can generalize to deploy in a physical environment.

22. **A Cooperative Keypoint–Sparse Cache and Improved PPO Framework for Rapid 3D UAV Path Planning**

    Abstrak :

    UAV path planning in complex 3D terrain faces the dual challenges of computational efficiency and reliable obstacle avoidance. To address these issues, this paper proposes a Keypoint–Sparse Cache (KSC) strategy and a hierarchical KSC-PPO (Proximal Policy Optimization) framework for mountainous environments with both static terrain and dynamic obstacles. The KSC strategy reduces search complexity through orthogonal slice-based sparse keypoint extraction and path caching reuse, thereby improving the efficiency of global path planning. On this basis, PPO-based local obstacle avoidance is activated only when safety thresholds are exceeded, while the remaining path is replanned globally after threat clearance, which confines avoidance computation to a local scope while preserving global path quality. Experiments in static mountainous environments show that KSC requires substantially less computation time than RRT* and Informed RRT* while maintaining competitive path efficiency, and it also outperforms four bio-inspired optimization algorithms across terrains of increasing complexity. Hybrid navigation validation experiments further show that KSC-PPO achieves high mission success, low collision rates, and low avoidance overhead in dynamic mountainous environments. Experiments demonstrate that KSC-PPO decomposes exponential global search space into controllable linear subproblems, significantly enhancing efficiency while ensuring path quality, providing an effective solution for UAV navigation in complex terrain.

23. **Hierarchical Path Planning based on PPO for UVs on 3D Off-Road Terrain**

    Abstrak :

    In 3D off-road terrain, the driving of the unmanned vehicle (UV) is influenced by the combined effect of terrain and obstacles, leading to greater challenges in safety and stability. Many path planning studies on 3D off-road terrain are lacking

in the consideration of this combined effect, for which we implemented a continuous simulation system and made the following contributions. First, an integrated potential field that is adaptive to the size of obstacles is proposed, and then it is used as a movement cost in the Voronoi diagram (VD) to obtain a pre-planned path for obstacle avoidance. Second, the Proximal Policy Optimization (PPO) algorithm is used to reduce the task objective, the effects of terrain and obstacles, and vehicle motion constraints to an objective function to be optimized. Finally, to improve the learning efficiency and generalization ability, we propose a hierarchical path planning (HPP) method to introduce the derived pre-planned path into the learning process of the PPO algorithm. It is experimentally demonstrated that the PPO algorithm combined with the HPP method is able to accomplish the path planning task in 3D off-road terrain of different sizes and difficulties, and obtains higher accuracy and shorter 3D path than the shaping reward (SR) method. The final path is suitable for the UV without smoothing, and the UV automatically adjusts its steering on undulating terrain to improve stability, and automatically slows down to pass through sharp turns to reduce the risk of rollover.

24. **Research on Autonomous Driving Route Planning Method Based on Deep Reinforcement Learning**

    Abstrak :

    This paper introduces a hybrid deep reinforcement learning path planning framework (DRL-PP) for autonomous vehicles. It is designed to be efficient, safe and verifiable in complex and dynamic environments. This framework is built on a hierarchical action representation: the policy network provides multi-step reference path points, and at the lower level, MPC/PID is used to ensure that things are physically possible. The reward function adopts a combinatorial design and incorporates distancebased cubic boosters to amplify the learning signal when approaching the target. The training strategy employs a dualnetwork setup, experience replay, and a cautious.

25. **Continuous Decision-Making for Risk-Based Collision Avoidance in Autonomous Driving using Twin-Delayed DDPG**

    Abstrak :

    The issue of collision avoidance (CA) represents a core challenge for the safe operation of autonomous driving vehicles. In this paper, we propose a novel CA method that integrates a risk evaluation strategy into deep reinforcement learning (DRL). The method formulates the CA problem in autonomous driving as a minimum risk optimization problem (MROP). To solve this formulated MROP, a risk evaluation model was first constructed to estimate the specific risk probabilities of a given driving state. Subsequently, the risk evaluation result was

incorporated into the reward function. The design of the DRL module and the comparison of multiple deep reinforcement learning algorithms for optimal CA policy were also conducted. Finally, the proposed method was tested on a traffic scenario with dense obstacles in the Carla environment. The experimental results demonstrated the efficacy of the proposed approach. Furthermore, the impact of certain DRL parameter settings on the learning of CA strategies was investigated.

26. Physics enhanced residual policy learning (PERPL) for safety cruising in mixed traffic platooning under actuator and communication delay
    Linear control models have gained extensive application in vehicle control due to their ease of use, and support for stability analysis. However, these models lack adaptability to the changing environment and multi-objective settings. Reinforcement learning (RL) models, on the other hand, offer adaptability but suffer from poor interpretability and safety issues that arise from limited out-of-distribution (OOD) generalization. This paper develops an RL-based controller enhanced by physics-informed policy, leveraging the advantages of both linear control models (data-efficient and interpretable) and RL models (flexible to multiple objectives and fast computing). We propose the Physics-Enhanced Residual Policy Learning (PERPL) framework, where the physics component provides model interpretability and stability. The learning-based Residual Policy adjusts the physics-based policy to adapt to the changing environment, thereby refining the decisions of the physics model. Theoretical analysis shows that, under local approximation conditions, PERPL can exhibit a smaller Lipschitz constant and less restrictive stability requirements compared to a pure RL policy. We apply the proposed model to decentralized cruising control in mixed traffic platoon of Connected and Automated Vehicles (CAVs) and Human-driven Vehicles (HVs) with actuator and communication delays. Experimental results demonstrate that PERPL achieves smaller headway errors and better oscillation dampening than linear models and RL in unseen scenarios with artificially extreme conditions. At the macroscopic level, overall traffic oscillations are also reduced as the penetration rate of CAVs employing the PERPL scheme increases, which matches the theoretical conclusion.